

LIBRARY MANAGEMENT SYSTEMS

A DISSERTATION

Presented to:

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THE PROBLEM

Introduction

Academic developers must provide evidence of the impact of our work. Whether we like the undercurrent of the word or not, Kathryn has written elsewhere about the choices we make with the language we use in academic development (see Sutherland, 2015). We should all aim to show that our work is meaningful, valuable, worth the time and effort invested in it. Philippine initiative to assimilate Education for Sustainable Development (ESD) in tertiary education is enunciated in the Commission on Higher Education's Strategic Plan for 2011–2016, which contains the roadmap to reform higher education institutions (HEIs) to respond well to the academic pursuits of sustainable development.

This study focuses on the issues faced in libraries. The Library Management System includes borrowing, returning books, classification of materials, cataloging, and barcoding. Barcode technology had its first commercial applications introduced in the late 1980s. Barcode uses electromagnetic fields to automatically identify and track tags attached to objects. A Barcode system consists of a tiny radio transponder, a radio receiver, and a transmitter. When triggered by an electromagnetic interrogation pulse from a nearby Barcode reader device, the tag transmits digital data, usually an identifying inventory number, back to the reader.

The goal of our research is to implement Barcode technology. It offers many significant benefits for library management and how these benefits reflect

students' and faculties satisfaction. This research helps the students for academic development to have a convenient way of accessing the library.

Statement of the Problem:

This study will support and manage library transactions and services in Opol Community College. Also, we will be able to develop a system using Laravel software for the design in library monitoring in response to the current manual library operations services, and transactions are enumerated below.

- **Lack of Progress-** Presently, the transaction of books in the institutional libraries has been done manually in most cases. It takes time for the transactions like borrowing or return of books. And specially to searching of borrowers and books.
- **Misplace Library Cards-** In most cases, inefficient library management, it may cause by human error because of loss and damages of records due to not using a computerized system in the library.
- **Time Consuming** - Difficulty in searching for books, the library services in terms of borrowed books through index cards consumed a lot of time.
- **Issuing of Books is Slow** - The issuing of books is slow, where borrowers (faculties, students, staff) will have to wait for further library transactions.

Objectives of the Study

General Objective

This study aimed to develop a manual library management system for fast transactions and to monitor, manage all library operations efficiently and easy to access data.

Specific Objectives

- a) To create automated Library Management with Bar Code Scanner
- b) To build a monitoring system that is able to monitor and manage all library operations efficiently.
- c) To give an opportunity to librarians to reduce mistakes that always happen during manual method.
- d) To store properly the library items in order to maintain their security.
- e) To build a system that can receive input and generate automatically output in easy way and short time.

Significance of the Study

The proposed system will inspire other students to develop an effective and efficient system. The proposed system will introduce technology to the school libraries that are until now adapting manual methods. The result of this study is beneficial to the Library will simplify and automate everyday tasks and can help maximize time spent with the borrowers thereby providing better service making it more profitable.

Librarian, will make it easier for the librarian to manage the library and convenient when it comes to retrieving book records. Students, will make it easier for the students to borrow books and to lessen their time in borrowing. Researcher, the researcher will develop their writing, analysis, and interpretation skills needed to make a good research. Future Reseacher, this will benefit other researchers who wish to have similar studies as they can get background information from the result of this study which will serve as a template to modify their research.

Definition of Key Terms

Users ID is a unique identifier, commonly used to log on to a website, app, or online service. It may be a username, account number, or email address.

Password a secret word or phrase that must be used to gain admission to something.

Email Address is a unique identifier for an email account. It is used to both send and receive email messages over the Internet.

Author is the creator or originator of any written work such as a book or play, and is also considered a writer.

ISBN is an International Standard Book Number. ISBNs are calculated using a specific mathematical formula and include a check digit to validate the number.

Book is a medium for recording information in the form of writing or images, typically composed of many pages (made of papyrus, parchment, vellum, or paper) bound together and protected by a cover.

Date of Publication the date on which a book or magazine is published, or is going to be published.

Borrower a person or organization that takes and uses something belonging to someone else with the intention of returning it.

Book Title the title of a book, or any other published text or work of art, is a name for the work which is usually chosen by the author.

Date Issue the date on which the Bonds are delivered.

Date Return The date on which a specified person has been notified to appear.

RESULT DISCUSSION

This section discusses four (4) phases by Dream Team in developing this Library Management System, namely the planning phase, requirements gathering phase, design phase, and development phase of the Library Management System with barcode scanner and its contribution to students' academic development.

The development of the system will be constrained by the availability of required software such as web servers, database and development tools. Library Management system can potentially have more than hundreds of users. It is unrealistic to provide training for everyone. The library management system should not have any unscheduled down time during library operation hours. Any down time in operation hours has significant impact to the operation and will cause inconvenience to everyone in library. If project constraints will be surpassed the operation inside the library will be efficient and it will be an environment that is conducive for learning.

Initial Planning

In this phase, All of us have a different task to develop the manual Library Management System by gathering information that we can apply to our system. Also, by exchanging thoughts, brainstorming, searching for other references. Every time our weekends are free, we meet up face to face to cope up and gather all information that we've assigned.

During this phase, We focus on the problem encountered in manual library management, what are the solutions to solve those problems. Especially in monitoring, borrowing, returning, loss and damages for the satisfaction of the user. We gather survey questionnaires for the student of Opol Community College to inform them and if they are aware of this Library Management System with Barcode Scanner.

Requirements Gathering

The Dream Team gathered some reports that use as a reference to describe what information that would be relevant to the solutions might help our proposed system. As shown in Table 1, below are the reports on using a manual library management system.

Table 1: Reports in using Manual Library Management System

Forms Name/Report Name	Description	Data/Information
Log Books	This form is to be filled up by the user once you enter the library.	Student Name Course and Year Time In Student ID

Index Card	This form is used by the user for borrowing and returning books.	Student Name Course and Year Book Title Author Date of Borrowing Date of Returning
List of Books	This form is a manual data monitoring of a librarian to input all book availability, which may cause loss and damages to data.	Book Title Book ID Book Name ISBN
Borrowing/Returning Reports	This form is manually used by the librarian to input data using log books.	Book ID Student ID Name Course Address Date Borrowing Date Returning

System Modules and Functional Requirements

The team created modules that describe their roles and functionality based on information gathered from the various Library Management reports.

Admin's Module. In this module, the admin will create an entry of a book, monitoring inventory. View all the transactions below is the detailed list of functionality for this module (see table 2).

Table 2: Functionality of Admin's Module

Functions	Description
Registration	
Submit	Adding new admin to the system which stores for each member identification number, name, address, street, city, phone number, age, and email address. Data is stored in a separate file in the same format of books file.
Login Form	
Login	Admin also has the authority to provide username and password for the receptionist.
Borrowing	
Add	For each member who borrows a book, the admin can view all the transactions like ISBN, user ID, date issued, date due to return, date returned. Also,
Edit	
Delete	

	the admin will manage the book Add Books, Edit, and Delete.
Returning	
Add	When a user returns a borrowed book to the library. The entry for his borrowing action must be updated with the data returned. Also for book entry, the number of available copies of the book must be incremented. Admin will also Add, Edit and, Delete.
Edit	
Delete	
Book List	
Search	When the admin wants to view all the books. Admin can search if he wants to check a book.
Returned List	
Search	The admin can view all the information that returned on that day. Admin can search if he wants to locate the name of the user who returned.
Borrowing Slip	
Print	The admin will print the borrowers slip. For receipt.
Books	
Add	Admin can also add the details of the book purchased from shops along with the shop name. He also has the authority to delete from or modify it to the database.
Edit	
Delete	
Books Category	
Add	Admin will manage all the available books by category. He can add another available issue of
Edit	
Delete	

	books, edit, and also delete books that are not available.
Borrowers Report	
Search	The admin will view all the report of borrowers, he can search also the names of the borrowers.
Penalty	
Add	If the borrowers are overdue or loss and damages of the books the admin will automatically inform.
Edit	
Delete	

Borrower Module. In this module, the user will register and log in for the students who want to borrow a book, and also it indicates the history of borrowed books and the availability of books. Table 3 are the lists of all the mentioned functionalities.

Table 3: This Functionality of Borrower's Module

Functions	Description
Registration	
Submit	Adding new borrowers to the system which stores for each member identification number, name, address, street, city, phone number, age, and email address.

Login Form	
Login	Borrower will login and provide Username and Password.
Borrowing	
Add	Borrowers can borrow books, They can even add their chosen book. Edit and Delete Books.
Edit	
Delete	
Returning	
Add	When a borrower return a borrowed book back to the library, the entry for his borrowing action must be updated with the date returned. Also for book entry; the number of available copies of the book must be incremented.
Edit	
Delete	
Book List	
Search	Allows the borrower to search any book given any combination of Book id and/or Book Title and /or Book Author.
Returned List	
Search	When the borrower wants to returned a book. The returned book will be input in the returned list.
Borrowing Slip	
Print	‘When the borrower have already borrow or returned a book their will be a receipt that indicates the borrowed and returned books has been borrow or returned.
Books	

Add	Borrower can easily add books on their preferred one and edit the one he adds and also delete .
Edit	
Delete	
Books Category	
Search	To search books by title, author, subject or publication date
Borrowers Report	
Search	To view all the borrowers report, they can search for easily access of report borrower.
Penalty	
Add	If the borrowers are overdue or loss and damages of the books.
Edit	
Delete	

Librarian Module. This module librarian can encode all the transactions of the day. Monitoring of books, confirming for borrowers, print receipt. Table 4.3 lists all the mentioned functionalities.

Table 4: Functionality Module for the librarian

Functions	Description
Registration	
Submit	Adding new librarian to the system which stores for each member identification number, name,

	address, street, city, phone number, age, and email address.
Login Form	
Login	Librarian will login and provide Username and Password.
Borrowing	
Add	Librarian manage all the transactions whom borrowed everyday, He can even add new borrower. Edit and Delete Borrower.
Edit	
Delete	
Returning	
Add	When a borrower return a borrowed book back to the library, The Librarian will automatically update the status of the borrower.
Edit	
Delete	
Book List	
Search	Librarian will search any book given any combination of Book id and/or Book Title and /or Book Author. Also the available books.
Returned List	
Search	When the borrower wants to returned a book. The librarian will automatically update the status of the borrower.
Borrowing Slip	

Print	When the borrower have already borrow or returned a book the librarian will give a receipt that indicates the borrowed books or returned books.
Books	
Add	Librarian can easily add books on their preferred one and edit the one he adds and also delete .
Edit	
Delete	
Books Category	
Search	To search books by title, author, subject or publication date
Borrowers Report	
Search	To view all the borrowers report, they can search for easily access of report borrower.
Penalty	
Add	If the borrowers are overdue or loss and damages of the books the librarian will give them a penalty.
Edit	
Delete	

Analysis and Design

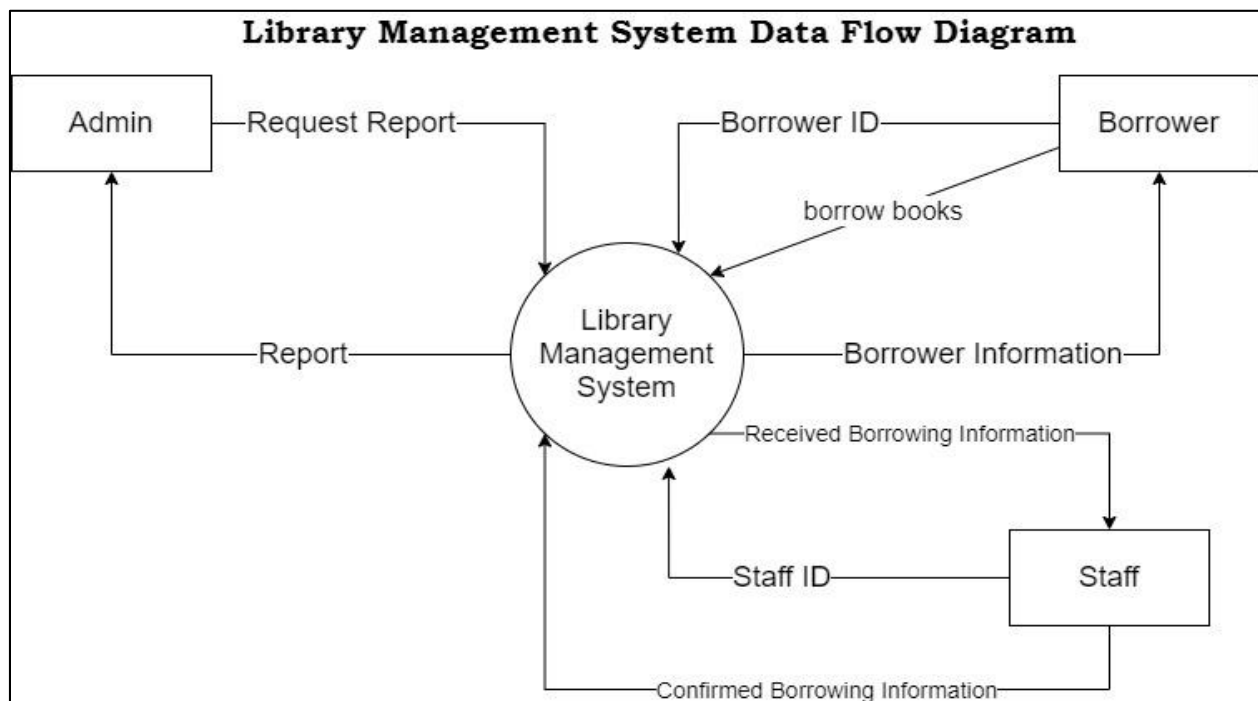
We could now translate all of the information obtained in the previous step during this step. It will also help to present the system's analysis and design, which helped in the deployment of the Library Management System.

Conceptual Diagram

As shown on the figure 1 below is the context diagram of Library Management System. This will give you an insight on what the Library Management is all about. The identified entities are the initiator where all the data flows in the system.

The context diagram of the LMS is presented in Figure 1. This will give you a better understanding of what Library Management entails. The specified entities are the system's initiators, via which all data flows.

Figure 1: System Data Flow Diagram of the LMS



Admin Entity

The admin can view all the module. It can generate all kinds of reports. Also can view the audit trail report of the logs activity of the borrower and the staff. The admin also can confirm the borrowed book from the borrower in the absence of staff (librarian). Admin can register the new list of books, category of books and also receive from new purchases of books. The admin has the authority to add, delete or update the details of the books from the system.

Borrower Entity

The borrower, it is a self registration of the LMS to input the personal information. This is a way of traicing the identity of the borrower. Can view the inventory of the books and select the book to be borrowed. If the borrower successfully transact in the borrowing form an

Staff

Also known as the librarian. A person in-charge of the library. The staff has the authority to confirm what the borrowers borrowed.

Conceptual Diagram Level 1

As shown on the figure 2 below is the Data Flow Diagram Level 1 of the Library Management System. This diagram will give you more insight on how the Library Management System works and store the data. It shows how each system works as the entity would initiate an action.

Figure 2.1: Data Flow Diagram Level 1

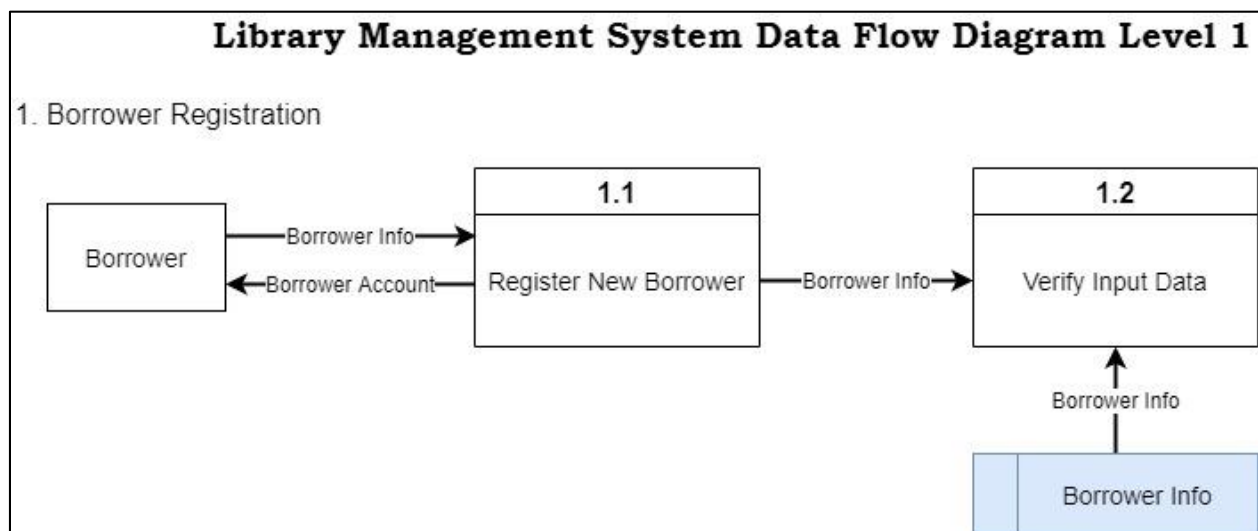


Figure 2.2: Data Flow Diagram Level 1

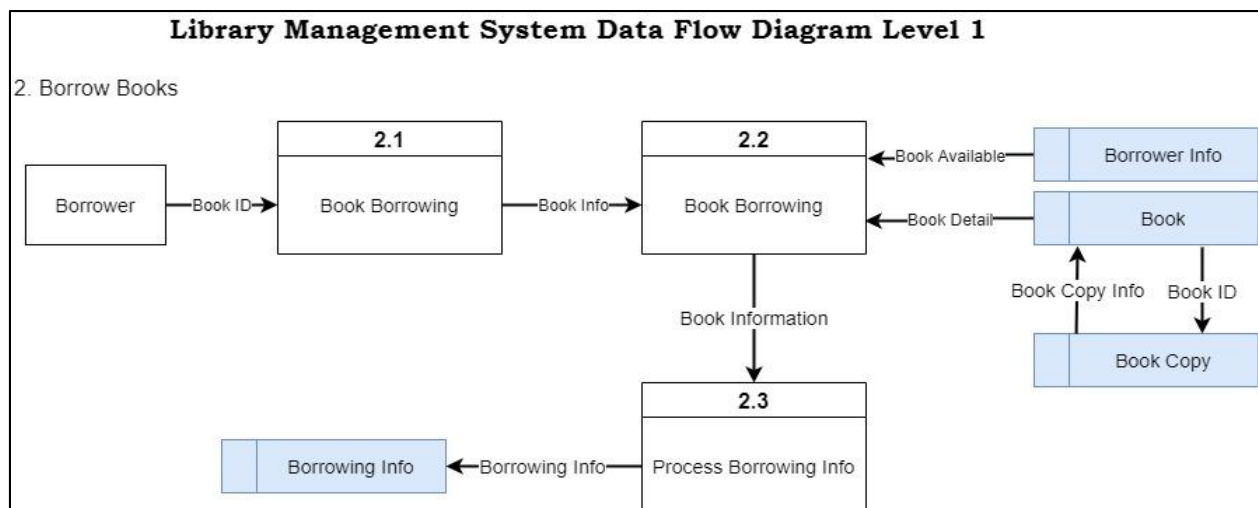


Figure 2.3: Data Flow Diagram Level 1

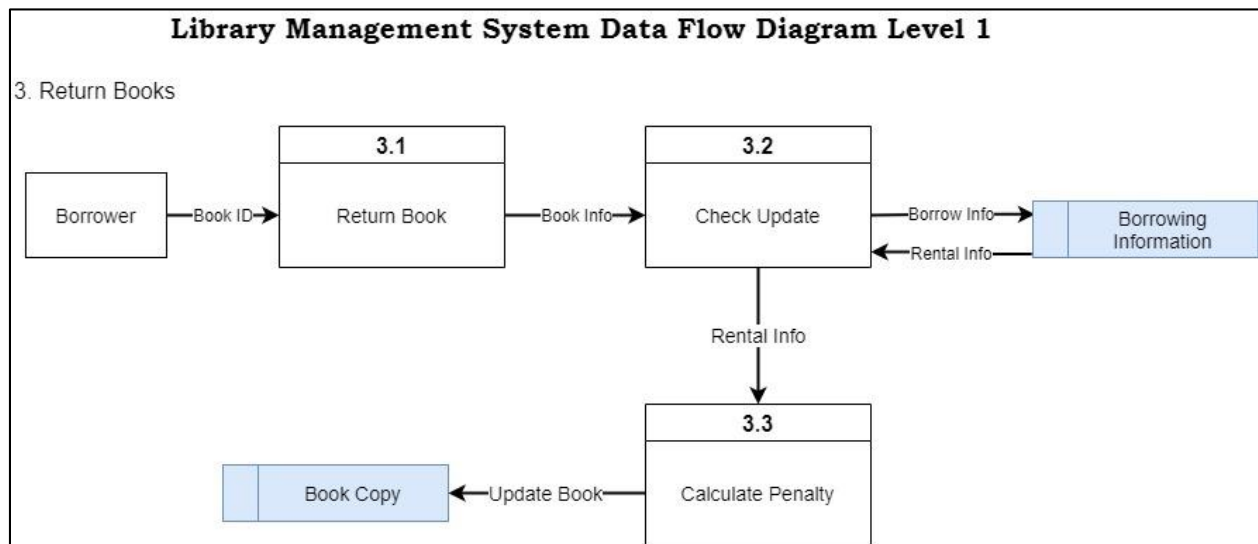
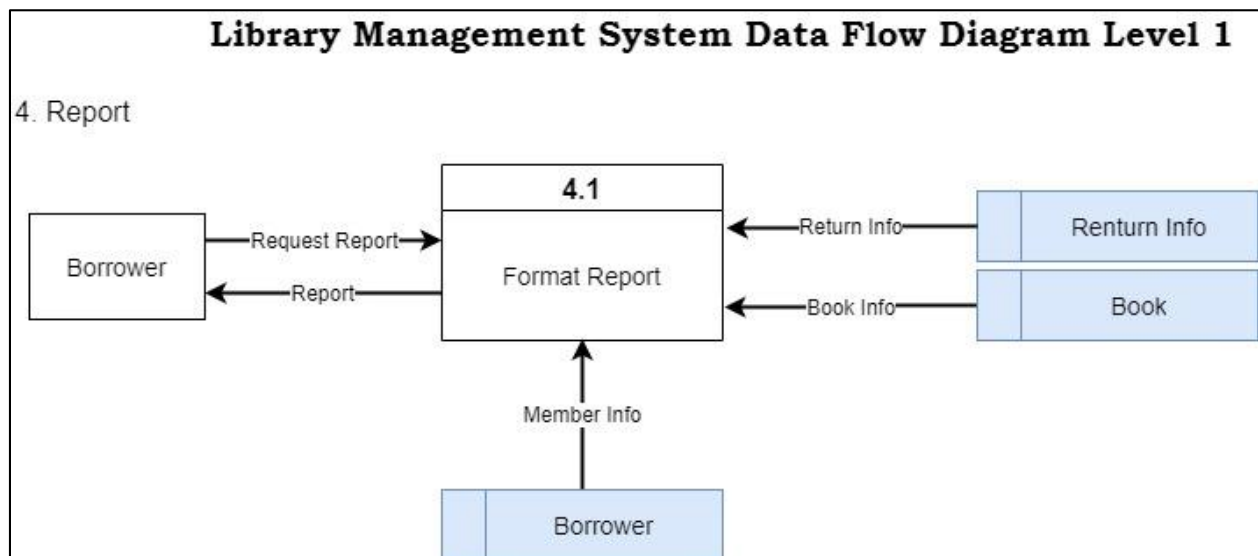


Figure 2.4: Data Flow Diagram Level 1



Borrower Registration

The borrower is the one who will register in the system. The borrower information will be save to the database. The system will generate a unique borrower id.

Borrow Book

The borrower is personally login their user account and open the borrow books module. The borrower can search and select the book they want to borrow and it will store the information to the database. The system will generate a unique borrowing id. The staff will confirm the borrow book and it will store to the database

Return Book

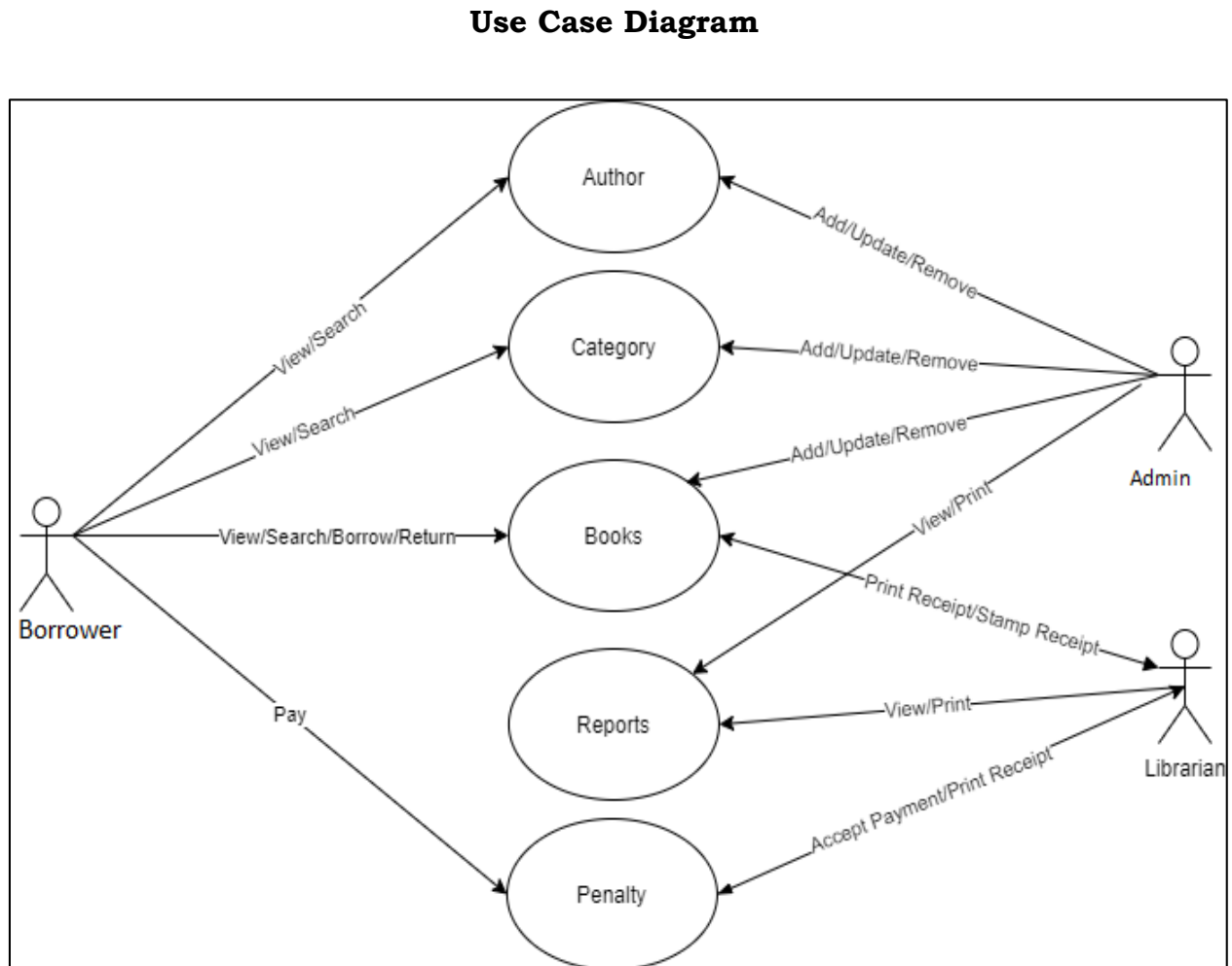
The borrower, login the system and select the return module then retrieve the borrow information using the borrowing id. The staff will confirm the return and the system will compute the penalty if they have.

Report

The Staff and/or the admin user can generate the report like the inventory of books, penalty report, borrowed/return books per day.

As shown on the figure 3 below is the illustration on how each Borrower, Librarian and Admin shows their roles and functions in the LMS

Figure 3: Use Case Diagram of the LMS.



Borrower

The borrower can search and view the author, category, book information, and penalty.

Admin

The admin can Add, Update, Delete the author, category, books, penalty, and can view all reports.

Staff (Librarian)

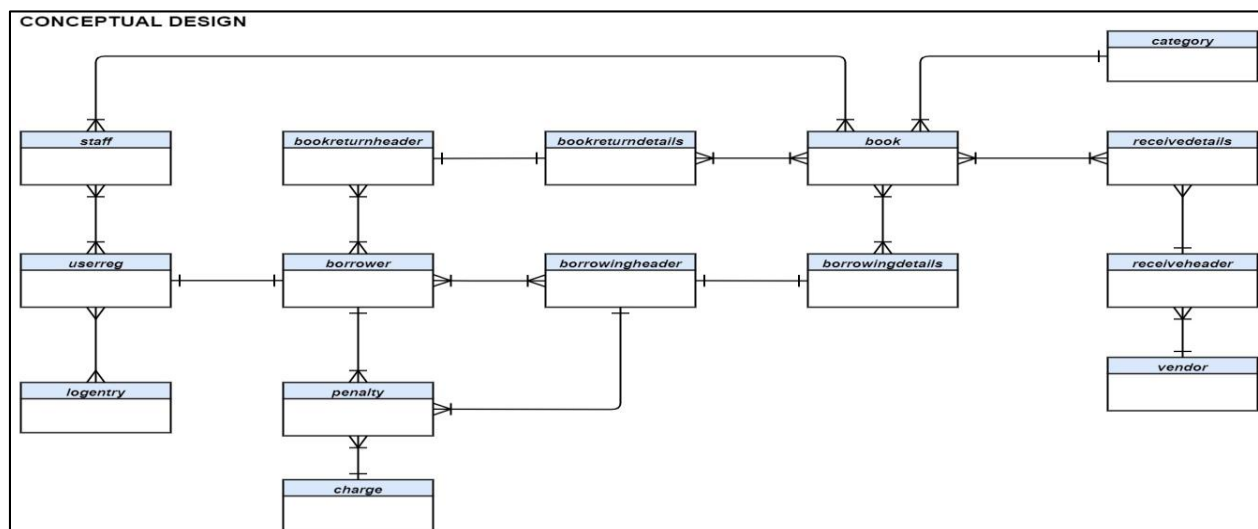
The staff will confirm the borrow and return of book and print a borrowers receipt. Can view all reports and accept penalty of the borrower.

Database Design

Conceptual Model Design

As shown on the figure 4 below, is the conceptual model design that represent a Library Management System. It is a composition of concepts which are used to help people know, understand or simulate how the system works.

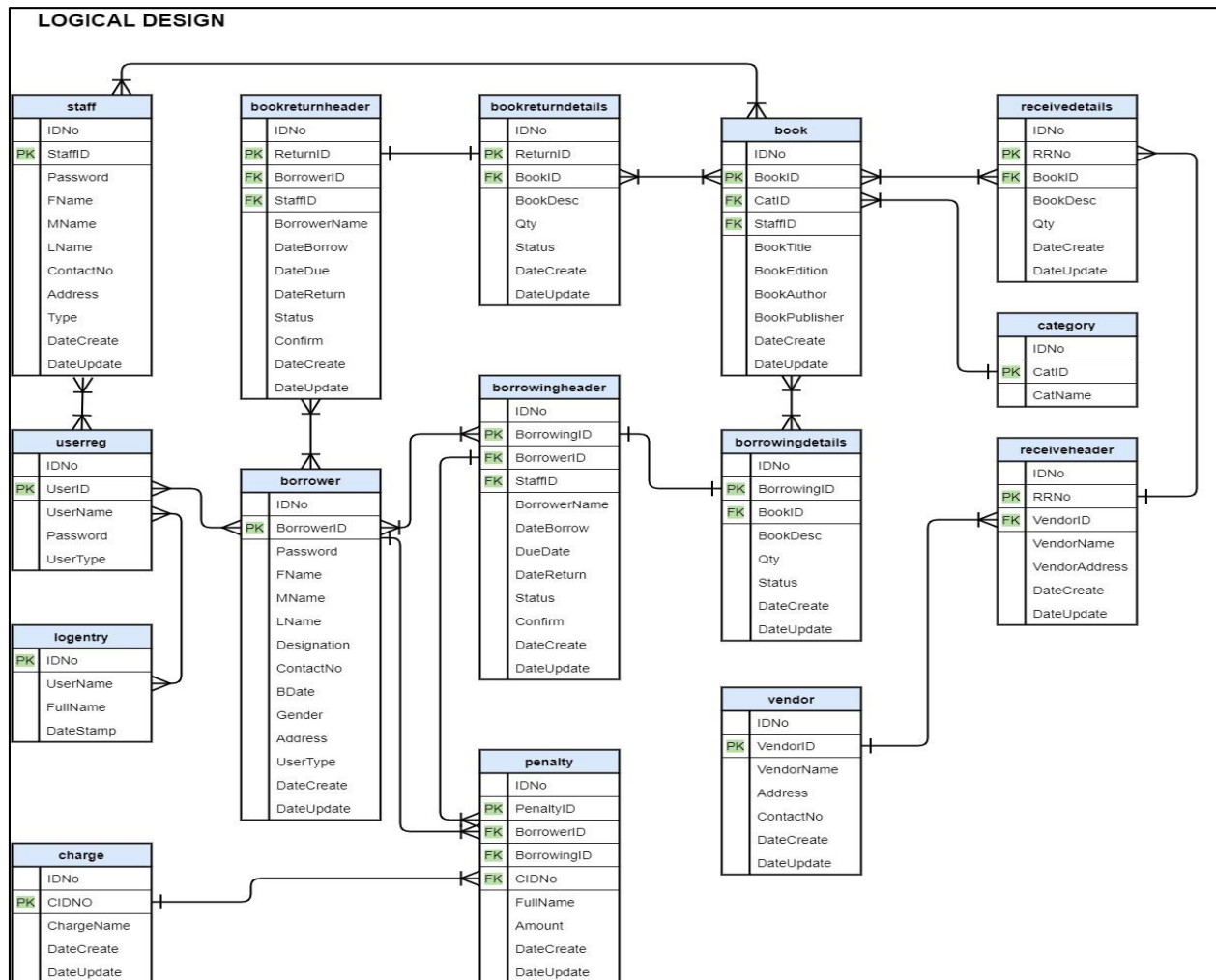
Figure 4: Conceptual Model of the Library Management System



Logical Model Design

As shown on the figure 5 below, is the logical model design that represents a Library Management System. It is a representation of each entity and their corresponding attributes

Figure 5: Logical Model of the Library Management System

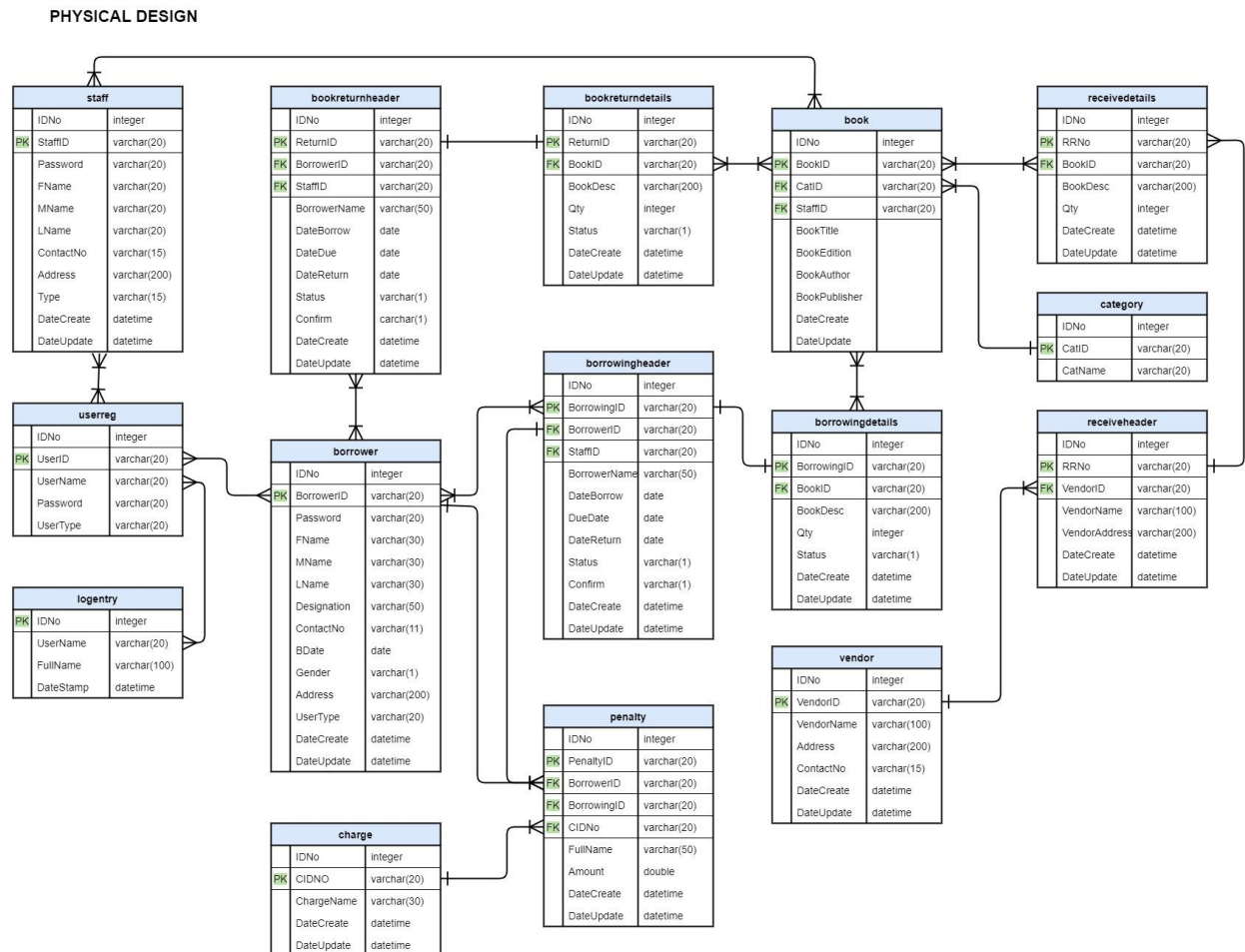


Each entity has corresponding attributes that constitute information of the Library Management System. It also reflects how each attributes of every entity associated with other attributes of other entity

Physical Model Design

As shown on the figure 6 below, is the physical model design that represent a Library Management System

Figure 6: Physical Model of the Library Management System



The physical model show how each attributes define the type and the length that was used. It also shows the relationship of each entity with the other entity. The figure shows the actual Library Management System.

Data Dictionary

Here's the list of different functions of our system. The Data Dictionary of the system shows the collection of names, definitions, and attributes about data elements that we used or captured in a Library Management System. It describes the meanings and purposes of data elements of the library management system and provides guidance on interpretation, accepted meaning, and representation.

Book Table: As shown in table 5, presents the different attributes of the Book's table.

Field Name	Field Size	Data Type	Description
IDNo	11	Integer,Auto_Increment	
BookID (PK)	20	varchar	Primary Key of book table
BookTitle	50	varchar	It's aimed to catch the reader's attention and reflect the narrator's perspective
BookEdition	20	varchar	Is a particular version of a book, magazine, or newspaper that is printed at one time
BookAuthor	200	varchar	is the creator or originator of any written work such as a book or play, and is also considered a writer
BookPublisher	100	varchar	a person or company whose business is the publishing of books, periodicals, engraving, computer software, etc. The business head of a newspaper organization or publishing house, commonly the owner or the representative of the owner.
CatID (FK)	20	varchar	Foreign key to Category table
StaffID (FK)	11	integer	Foreign key to StaffID table

DateCreate		datetime	DateCreated
DateUpdate		datetime	DateUpdated

BookReturnDetails Table: As shown in table 6, presents the different attributes of the Book Return Details table.

Field Name	Field Size	Data Type	Description
IDNo	11	Integer,Auto_Increment	
ReturnID (PK)	20	varchar	Primary key of borrowingdetails table
BookID (FK)	20	varchar	Foreign key to book table
BookDesc	200	varchar	Book Title / Book Edition / Book Author / Book Publisher FROM Category Table
Qty	11	Integer	Number of book/s borrowed
Status	1	varchar	Status of borrowed book O/C (Open/Close)
DataCreate		datetime	Date created
DataUpdate		datetime	Date updated

BookReturnHeader Table: As shown in table 7, presents the different attributes of the Book Return Header table.

Field Name	Field Size	Data Type	Description
IDNo	11	Integer,Auto_Increment	
ReturnID (PK)	20	varchar	Primary key of bookreturnheader table
BorrowerID (FK)	20	varchar	Foreign key to borrower table
BorrowerName	50	varchar	Name of borrower
StaffID (FK)	20	varchar	Foreign key to staff table
DateBorrow		date	date when book is borrowed (YYYY-MM-DD)
DateDue		date	Due date of book borrowed (YYYY-MM-DD)
DateReturn		date	Date when book is returned (YYYY-MM-DD)
Status	1	varchar	Status of borrowed book O/C (Open/Close)
Confirm	1	varchar	Confirmed by Staff of library Y/N (Yes/No)
DataCreate		datetime	Date created
DataUpdate		datetime	Date updated

Borrower Table: As shown in table 8, presents the different attributes of the Borrower's table.

Field Name	Field Size	Data Type	Description
IDNo	11	Integer,Auto_Increment	
BorrowerID (PK)	20	varchar	Primary key for borrower table
Password	20	varchar	Password for login
FName	30	varchar	Name of borrower
MName	30	varchar	Middle name of borrower
LName	30	varchar	Last name of borrower
Designation	30	varchar	Student/Employee/Instructor

ContactNO	11	varchar	Contact number of borrower
Bdate		Date	Birthday of borrower (YYYY-MM-DD)
Gender	1	varchar	Gender of borrower M/F (Male/female)
Address	200	varchar	Address of borrower
UserType	20	varchar	Type of user (Admin/librarian/borrower)
DateCreate		Datetime	Date borrower is created
DateUpdate		Datetime	Date borrower is updated

BorrowingDetails Table: As shown in table 9, presents the different attributes of the Borrowing Details table.

Field Name	Field Size	Data Type	Description
IDNo	11	Integer,Auto_Increment	
BorrowingID (PK)	20	varchar	Primary key of borrowingdetails table
BookID (FK)	20	varchar	Foreign key to book table
BookDesc	200	varchar	Book Title / Book Edition / Book Author / Book Publisher FROM Category Table
Qty	11	Integer	Number og book/s borrowed
Status	1	varchar	Status of borrowed book O/C (Open/Close)
DataCreate		datetime	Date Borrowing created
DataUpdate		datetime	Date Borrowing updated

Category Table: As shown in table 10, presents the different attributes of the Category table.

Field Name	Field Size	Data Type	Description
IDNo	11	Integer,Auto_Increment	
CatID (PK)	10	varchar	Primary Key of book table
CatName	20	varchar	Category Name
DataCreate		datetime	Date Borrowing created
DataUpdate		datetime	Date Borrowing updated

Charge Table: As shown in table 11, presents the different attributes of the Charge's table.

Field Name	Field Size	Data Type	Description
IDNo	11	Integer,Auto_Increment	
CIDNo (PK)	20	varchar	Primary Key of Charge Table
ChargeName	20	varchar	Charge Name
Amount		double	Charge Amount (#,##0.00)
DataCreate		datetime	Date Borrowing created
DataUpdate		datetime	Date Borrowing updated

LogEntry Table: As shown in table 12, presents the different attributes of the Log Entry table.

Field Name	Field Size	Data Type	Description
IDNo	11	Integer,Auto_Increment	
UserName	20	varchar	User Name from UserReg Table
FullName	100	varchar	Full Name from Staff Table
DateStamp		datetime	Date & Time when the user login

Penalty Table: As shown in table 13, presents the different attributes of the Penalty table.

Field Name	Field Size	Data Type	Description
IDNo	11	Integer,Auto_Increment	
PenaltyID (PK)	20	varchar	Primary Key of Penalty Table
BorrowerID (FK)	20	varchar	Foreign Key from borrower Table
ReturnID (FK)	20	varchar	Foreign Key from bookreturnheader Table
FullName	50	varchar	BorrowerName from bookreturnheader Table
CIDNo (FK)	20	varchar	Foreign Key from Charge Table
Amount		double	Amount from Charge Table
DataCreate		datetime	Date Borrowing created
DataUpdate		datetime	Date Borrowing updated

ReceiveDetails Table: As shown in table 14, presents the different attributes of the Receive Details table.

Field Name	Field Size	Data Type	Description
IDNo		Integer,Auto_Increment	
RRNo (PK)	20	varchar	Primary Key of receivedetails Table
BookID (FK)	20	varchar	Foreign Key from book table
BookDesc	200	varchar	Book Title / Book Edition / Book Author / Book Publisher
Qty	11	integer	No. of book to receive from a vendor
DataCreate		datetime	Date Borrowing created
DataUpdate		datetime	Date Borrowing updated

ReceiveHeader Table: As shown in table 15, presents the different attributes of the Receive Header table.

Field Name	Field Size	Data Type	Description
IDNo	11	Integer,Auto_Increment	
RRNo (PK)	20	varchar	Primary Key of receiveheader
VendorID (FK)	20	varchar	Foreign Key of vendor table
VendorName	100	varchar	Vendor Name from vendor table
VendorAddress	200	varchar	Vendor Address from vendor table
DataCreate		datetime	Date Borrowing created
DataUpdate		datetime	Date Borrowing updated

Staff Table: As shown in table 16, presents the different attributes of the Staff table.

Field Name	Field Size	Data Type	Description
IDNo	11	Integer,Auto_Increment	
StaffID	20	varchar	Primary Key of staff table
Password	20	varchar	Password for login
Fname	20	varchar	Staff name
Mname	20	varchar	Staff middle name
Lname	20	varchar	Staff last name
ContactNo	15	varchar	Contact number of staff
Address	200	varchar	Address of staff
Type	15	varchar	Type of user
DataCreate		datetime	Date Borrowing created
DataUpdate		datetime	Date Borrowing updated

UserReg Table: As shown in table 17, presents the different attributes of the User Registration table.

Field Name	Field Size	Data Type	Description
IDNo	11	Integer,Auto_Increment	
UserID	30	varchar	Primary key of userreg table
UserName	20	varchar	Username from Staff ID and Borrower ID
Password	20	varchar	Password for login
UserType	20	varchar	Type of user

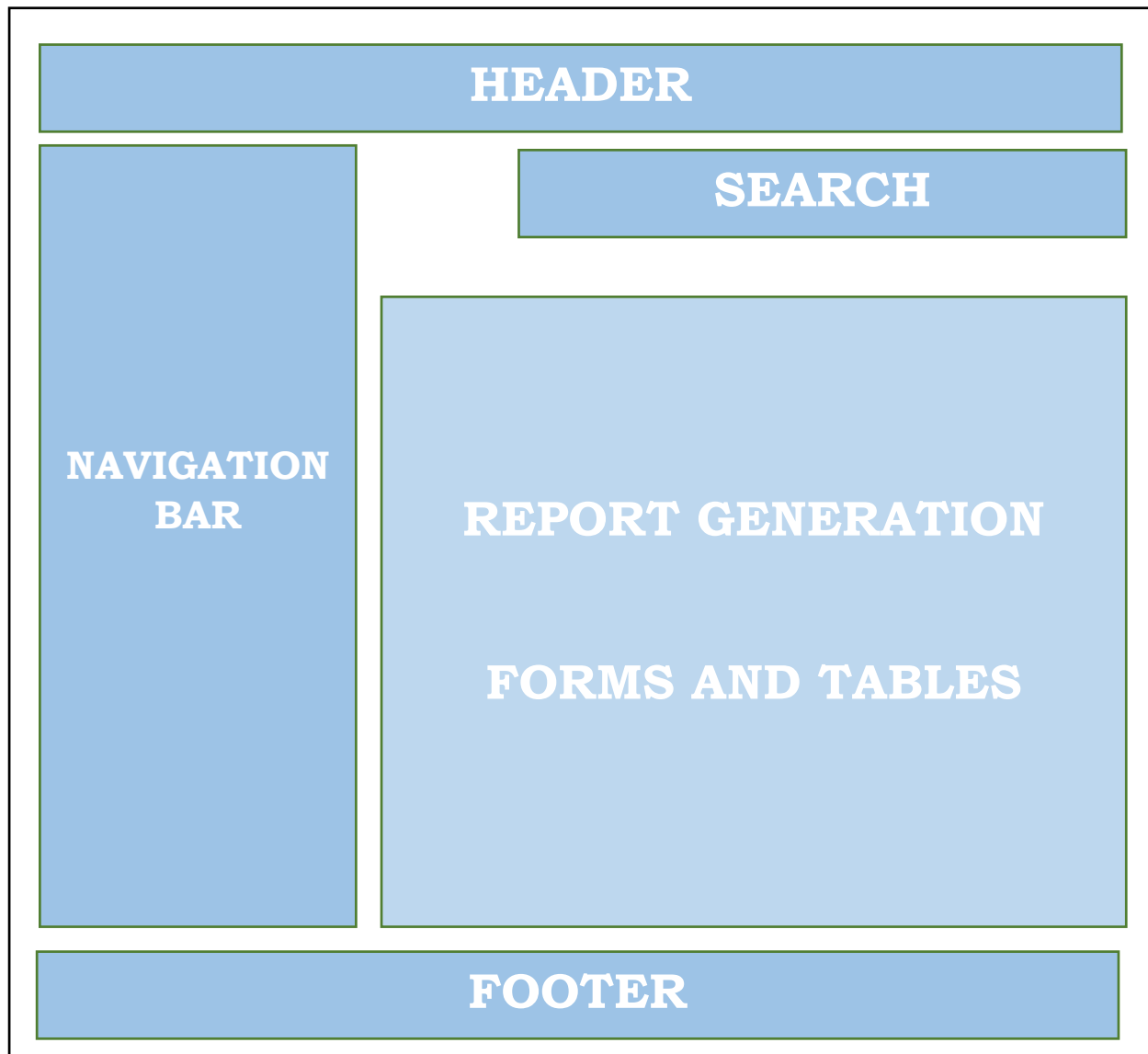
Vendor Table: As shown in table 18, presents the different attributes of the Vendor table.

Field Name	Field Size	Data Type	Description
IDNo	11	Integer,Auto_Increment	
VendorID (PK)	20	varchar	Primary key of vendor table
VendorName	100	varchar	Vendor Name
Address	200	varchar	Vendor Address
ContactNo	15	varchar	Vendor Contact Number
DataCreate		datetime	Date Borrowing created
DataUpdate		datetime	Date Borrowing updated

LO-FI Designs.

As shown of figure 7 is the LoFi design of the Library Management System.

Figure 7: LO_FI Design of the Library Management System.



Hi-Fi Design

Manage Books: As shown of figure 8 is the Hi-Fi design of the the Library Management System for the user interface with a query section.

Figure 8 : Hi-Fi Design on message interface of the Library Management System.

The image displays a web application interface for a Library Management System. The header is blue with a logo on the left, the title "Library Management System" in the center, and a search bar on the right. A left sidebar contains navigation links: Dashboard, Books, Borrowed/Returned, Add Books, Category, Borrower's Records, and Penalty. A "Logout" button is also present in the sidebar. The main content area is titled "MANAGE BOOKS" and features a form with nine input fields arranged in a 3x3 grid. Below the form is a table with columns for BOOK ID, BOOK Title, BOOK Edition, BOOK Publisher, BOOK Author, BOOK Category, No of Copies, No of Borrowed, and No of Available. The table contains five rows of data. At the bottom of the table are four orange buttons: Add, Edit, Delete, and Close. The footer is blue and contains the BSIT logo, copyright information, and links to the App Store and Google Play.

Library Management System

Search...

MANAGE BOOKS

Book ID Book Title Book Edition Book Publisher Book Author Book Category No of Copies No of Borrowed No of Available

1	Mark	Otto	@mdo	@mdo				
2	Jacob	Thornton	@fat	@mdo				
3	Larry the Bird	@twitter	@mdo	@mdo				
4	Mark	Otto	@mdo	@mdo				
5	Mark	Otto	@mdo	@mdo				

Add Edit Delete Close

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Design in Borrowing / Returning of Books: As shown of figure 9 is the Hi-Fi design of the the Library Management System for the user interface with a query section.

Figure 9: Hi-Fi Design in Borrowing/ Returning of books in Library Management System.

Library Management System

Dashboard

- Books
- Borrowed/Returned
- Add Books
- Category
- Borrower's Records
- Penalty

Logout

BORROWING/RETURNING OF BOOKS

BORROWING ID	BOOK ID	BOOK DESCRIPTION	QTY	ACTION
1	Mark	Otto	@mdo	@mdo
2	Jacob	Thornton	@fat	@mdo
3	Larry the Bird	@twitter	@mdo	@mdo
4	Mark	Otto	@mdo	@mdo
5	Mark	Otto	@mdo	@mdo

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Penalty: As shown of figure 10 is the Hi-Fi design of the the Library Management System for the user interface with a query section.

Figure 10: Hi-Fi Design in Penalty in Library Management System.

Library Management System

Search...

PENALTY

Penalty ID: Borrower's ID: Full Name:

Penalty Type: Amount: Borrowing ID:

PENALTY ID	BORROWER'S ID	FULL NAME	PENALTY TYPE	AMOUNT	BORROWING ID
1	Mark	Otto	@mdo	@mdo	
2	Jacob	Thornton	@fat	@mdo	
3	Larry the Bird	@twitter	@mdo	@mdo	
4	Mark	Otto	@mdo	@mdo	
5	Mark	Otto	@mdo	@mdo	

Add Edit Delete Close

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SUMMARY, CONCLUSION, AND RECOMMENDATION

This is the general summary, conclusion and recommendation of the project:
Library Management System.

Summary

Library Management System is a project which aims in developing a computerized system to maintain all the daily work of the library. This project has many features which generally not available in normal library management systems like facility of user login.

It also has a facility of admin login through which the admin can monitor the whole system. It has also facility where student after logging in their accounts can see list of books issued and its issue date and return date and also the students can request the librarian to add new books by filling the book request form.

The librarian after logging into his account the admin account can generate various reports such as borrowers report, issue report, book report. Overall this project of ours is being developed to help the students as well as staff of library to maintain the library in the best way possible and also reduce human efforts.

Conclusion

Library Management System is for computerizing the working in a library. The software takes care of all the requirements of a library and is capable to provide easy and effective storage of information related to books & users. It was an effort to develop a library management system that may be useful in a library.

This project will provide a computerized version of a library management system that will benefit the students as well as the staff of the library. It makes the entire process online where a student can search books, staff and admin can generate reports and do book transactions. It also has a facility for student login where the student can log in can log in and can see the status of books issued.

Recommendation

There is a need for people to take initiative to be computer literate in order to familiarize with these technologies and appreciate them. Library staffs need to undergo training in order to acquire basic skills in information technology which will enable them to use the library managements system. Libraries should embrace the trend of automating their services. This system should be extended to incorporate other modules for example the acquisition module among others, in order to be fully integrated.

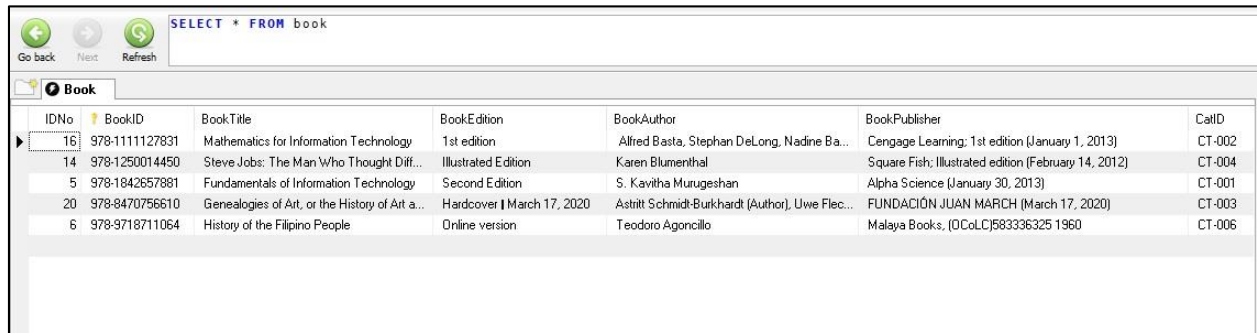
Future Work

In the near future, technology is growing rapidly and so this “Library Management Sytem” will be very helpful in the future. Easy to access and no loss of time. In the future, users can read online books like .pdf from library just by logging there account. In future, this system can launch website for eady online registration.

APPENDIX A

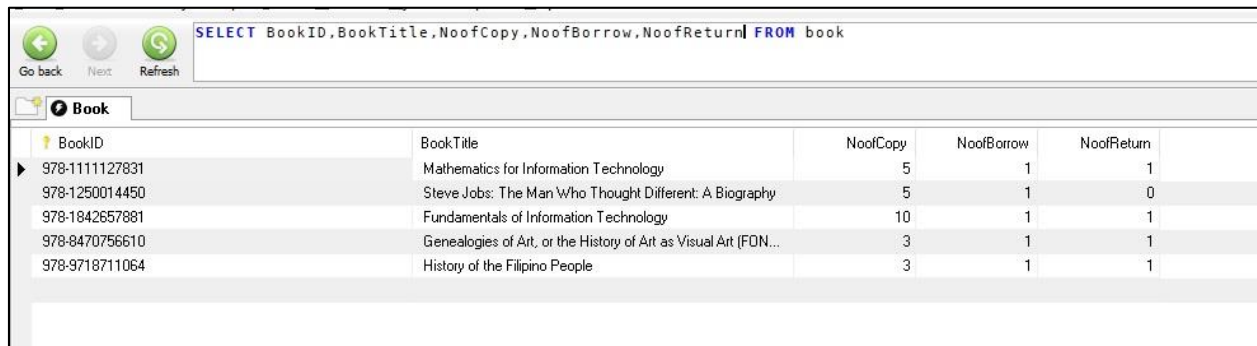
SAMPLE QUERIES

Figure 11: Sample query from the list of books.



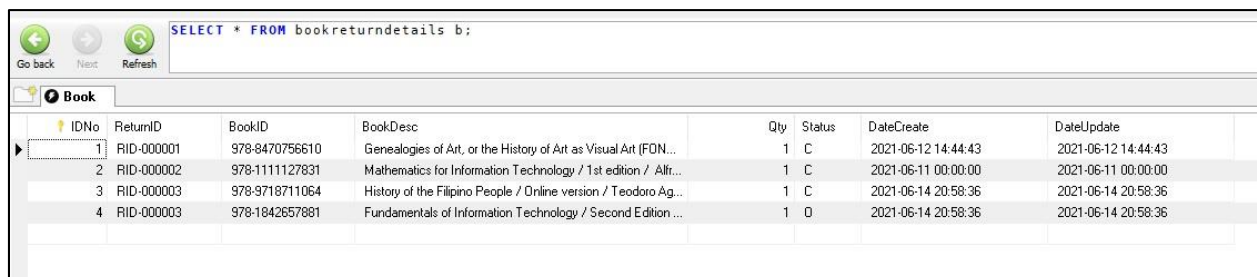
IDNo	BookID	BookTitle	BookEdition	BookAuthor	BookPublisher	CatID
16	978-1111127831	Mathematics for Information Technology	1st edition	Alfred Basta, Stephan DeLong, Nadine Ba...	Cengage Learning: 1st edition (January 1, 2013)	CT-002
14	978-1250014450	Steve Jobs: The Man Who Thought Diff...	Illustrated Edition	Karen Blumenthal	Square Fish; Illustrated edition (February 14, 2012)	CT-004
5	978-1842657881	Fundamentals of Information Technology	Second Edition	S. Kavitha Murugesan	Alpha Science (January 30, 2013)	CT-001
20	978-8470756610	Genealogies of Art, or the History of Art a...	Hardcover March 17, 2020	Astritt Schmidt-Burkhardt (Author), Uwe Flec...	FUNDACIÓN JUAN MARCH (March 17, 2020)	CT-003
6	978-9718711064	History of the Filipino People	Online version	Teodoro Agoncillo	Malaya Books, [OCoLC]583336325 1960	CT-006

Figure 12: Sample query from the list of books with selected table fields.



BookID	BookTitle	NoofCopy	NoofBorrow	NoofReturn
978-1111127831	Mathematics for Information Technology	5	1	1
978-1250014450	Steve Jobs: The Man Who Thought Different: A Biography	5	1	0
978-1842657881	Fundamentals of Information Technology	10	1	1
978-8470756610	Genealogies of Art, or the History of Art as Visual Art (FON...	3	1	1
978-9718711064	History of the Filipino People	3	1	1

Figure 13: Sample query from book return details



IDNo	ReturnID	BookID	BookDesc	Qty	Status	DateCreate	DateUpdate
1	RID-000001	978-8470756610	Genealogies of Art, or the History of Art as Visual Art (FON...	1	C	2021-06-12 14:44:43	2021-06-12 14:44:43
2	RID-000002	978-1111127831	Mathematics for Information Technology / 1st edition / Alfr...	1	C	2021-06-11 00:00:00	2021-06-11 00:00:00
3	RID-000003	978-9718711064	History of the Filipino People / Online version / Teodoro Ag...	1	C	2021-06-14 20:58:36	2021-06-14 20:58:36
4	RID-000003	978-1842657881	Fundamentals of Information Technology / Second Edition ...	1	O	2021-06-14 20:58:36	2021-06-14 20:58:36

Figure 14: Sample query from book return details where status is close

Go back

Next

Refresh

SELECT * FROM bookreturndetails where Status = 'C'

Book

IDNo	ReturnID	BookID	BookDesc	Qty	Status	DateCreate	DateUpdate
1	RID-000001	978-8470756610	Genealogies of Art, or the History of Art as Visual Art (FONDACI...	1	C	2021-06-12 14:44:43	2021-06-12 14:44:43
2	RID-000002	978-1111127831	Mathematics for Information Technology / 1st edition / Alfred B...	1	C	2021-06-11 00:00:00	2021-06-11 00:00:00
3	RID-000003	978-9718711064	History of the Filipino People / Online version / Teodoro Agoncill...	1	C	2021-06-14 20:58:36	2021-06-14 20:58:36

Figure 15: Sample query from book return details where status is open

Go back

Next

Refresh

SELECT * FROM bookreturndetails where Status = 'O'

Book

IDNo	ReturnID	BookID	BookDesc	Qty	Status	DateCreate	DateUpdate
4	RID-000003	978-1842657881	Fundamentals of Information Technology / Second Edition ...	1	O	2021-06-14 20:58:36	2021-06-14 20:58:36

Figure 16: Sample query from borrower list

Go back

Next

Refresh

SELECT * FROM borrower

Book

IDNo	BorrowerID	Password	FName	MName	LName	Designation	ContactNo	BDate	Gender	Address	UserType	DateCreate
1	2018-1-00100	Password1234	John	Te	Amo	Student	09123456789	1990-10-28	M	Opol, Misamis Oriental	Borrower	2021-06-01 1
2	2018-1-00101	Pass123	Dave	M.	Duriga	Student	09790989809	1990-06-01	M	El Salvador City	Borrower	2021-06-01 1
3	2018-1-00102	1234Pass	Stephanie Grace	Fisher	Namocot	Student	097909099	1995-07-01	F	Opol, Misamis Oriental	Borrower	2021-06-01 1
4	2018-1-00103	qwerty1234	John Warren	A.	Jaudian	Student	097909890	1992-11-28	M	Opol, Misamis Oriental	Borrower	2021-06-01 1
5	2018-1-00104	idontknow	Jumar	Pahayahay	Amores	Student	09145677888	1990-08-29	M	Iponan, Cagayan de Oro City	Borrower	2021-06-01 1

Figure 16: Sample query from borrowing details list

Go back

Next

Refresh

SELECT * FROM borrowingdetails bl;

Book

IDNo	BorrowingID	BookID	BookDesc	Qty	Status	DateCreate	DateUpdate
2	BID-000001	978-8470756610	Genealogies of Art, or the History of Art as Visual Art (FONDACION JUAN) / Hardcover March...	1	O	2021-06-03 20:12:31	2021-06-03 20:12:31
5	BID-000002	978-1111127831	Mathematics for Information Technology / 1st edition / Alfred Basta, Stephan DeLong, Nadine...	1	C	2021-06-03 20:23:43	2021-06-03 20:23:43
6	BID-000002	978-1250014450	Steve Jobs: The Man Who Thought Different: A Biography / Illustrated Edition / Karen Blument...	1	O	2021-06-03 20:23:43	2021-06-03 20:23:43
7	BID-000003	978-9718711064	History of the Filipino People / Online version / Teodoro Agoncillo / Malaya Books, [DCoLC]58...	1	C	2021-06-08 20:30:10	2021-06-08 20:30:10
8	BID-000003	978-1842657881	Fundamentals of Information Technology / Second Edition / S. Kavitha Murugesan / Alpha S...	1	C	2021-06-08 20:30:10	2021-06-08 20:30:10

Figure 17: Sample query from borrowing details where status is close



Go back



Next



Refresh

SELECT * FROM borrowingdetails where Status = 'C'



Book

IDNo	BorrowingID	BookID	BookDesc	Qty	Status	DateCreate	DateUpdate
5	BID-000002	978-1111127831	Mathematics for Information Technology / 1st edition / Alfred Basta, Stephan DeLo...	1	C	2021-06-03 20:23:43	2021-06-03 20:23:43
7	BID-000003	978-9718711064	History of the Filipino People / Online version / Teodoro Agoncillo / Malaya Books, (...)	1	C	2021-06-08 20:30:10	2021-06-08 20:30:10
8	BID-000003	978-1842657881	Fundamentals of Information Technology / Second Edition / S. Kavitha Murugesan...	1	C	2021-06-08 20:30:10	2021-06-08 20:30:10

Figure 18: Sample query from borrowing details where status is open







Go back

Next

Refresh

SELECT * FROM borrowingdetails where Status = 'O'



Book

IDNo	BorrowingID	BookID	BookDesc	Qty	Status	DateCreate	DateUpdate
2	BID-000001	978-8470756610	Genealogies of Art, or the History of Art as Visual Art (FONDACION JUAN) / Hardco...	1	O	2021-06-03 20:12:31	2021-06-03 20:12:31
6	BID-000002	978-1250014450	Steve Jobs: The Man Who Thought Different: A Biography / Illustrated Edition / Kar...	1	O	2021-06-03 20:23:43	2021-06-03 20:23:43

Figure 19: Sample query from borrowing header list

Go back

Next

Refresh

SELECT * FROM borrowingheader

Book

IDNo	BorrowingID	BorrowerID	BorrowerName	StaffID	DateBorrow	DueDate	DateReturn	Status	Confirm	DateCreate	DateUpdate
3	BID-000001	2018-1-00100	John Te Amo	2021-0-00001	2021-06-03	2021-06-07	2021-06-07	C	N	2021-06-03 20:12:36	2021-06-03 20:12:36
4	BID-000002	2018-1-00101	Dave M. Duriga	2021-0-00001	2021-06-03	2021-06-07	2021-06-07	C	N	2021-06-03 20:19:11	2021-06-03 20:19:11
5	BID-000003	2018-1-00102	John Warren A. Jaudian	2021-0-00001	2021-06-08	2021-06-11	2021-06-11	O	N	2021-06-08 20:30:10	2021-06-08 20:30:10

Figure 20: Sample query from borrowing header where status is close

Go back

Next

Refresh

SELECT * FROM borrowingheader where Status = 'C'

Book

IDNo	BorrowingID	BorrowerID	BorrowerName	StaffID	DateBorrow	DueDate	DateReturn	Status	Confirm	DateCreate	DateUpdate
3	BID-000001	2018-1-00100	John Te Amo	2021-0-00001	2021-06-03	2021-06-07	2021-06-07	C	N	2021-06-03 20:12:36	2021-06-03 20:12:36
4	BID-000002	2018-1-00101	Dave M. Duriga	2021-0-00001	2021-06-03	2021-06-07	2021-06-07	C	N	2021-06-03 20:19:11	2021-06-03 20:19:11

Figure 21: Sample query from borrowing header where status is open

Go back

Next

Refresh

SELECT * FROM borrowingheader where Status = 'O'

Book

IDNo	 BorrowingID	BorrowerID	BorrowerName	StaffID	DateBorrow	DueDate	DateReturn	Status	Confirm	DateCreate	DateUpdate
5	BID-000003	2018-1-00102	John Warren A. Jaudian	2021-0-00001	2021-06-08	2021-06-11	2021-06-11	O	N	2021-06-08 20:30:10	2021-06-08 20:30:10

Figure 22: Sample query from category list

Go back

Next

Refresh

SELECT * FROM category

Book

IDNo	CatID	CatName	DataCreate	DataUpdate
1	CT-001	Computer & Tech	2021-06-12 10:08:59	2021-06-12 10:08:59
2	CT-002	Science & Math	2021-06-01 16:12:06	2021-06-01 16:12:06
3	CT-003	Art & Music	2021-06-01 16:12:06	2021-06-01 16:12:06
4	CT-004	Biographies	2021-06-01 16:13:26	2021-06-01 16:13:26
5	CT-005	Business	2021-06-01 16:13:26	2021-06-01 16:13:26
6	CT-006	History	2021-06-01 16:14:15	2021-06-01 16:14:15
7	CT-007	Literature & Fiction	2021-06-01 16:14:15	2021-06-01 16:14:15

Figure 23: Sample query from charge list

Go back

Next

Refresh

SELECT * FROM charge

Book

IDNo	CIDNO	ChargeName	Amount	DateCreate	DateUpdate
1	CH-001	Late Return	10	2021-06-01 16:15:27	2021-06-01 16:15:27
2	CH-002	Damage	300	2021-06-01 16:15:27	2021-06-01 16:15:27
3	CH-003	Lost	500	2021-06-01 16:17:35	2021-06-01 16:17:35

Figure 24: Sample query from penalty list



Go back



Next



Refresh

SELECT * FROM penalty p;



Book

IDNo	PenaltyID	BorrowerID	ReturnID	DTPenalty	FullName	CIDNo	Amount	DateCreate	DateUpdate
1	PID-00001	2018-1-00103	RID-000003	2021-06-14	John Warren A. Jaudian	CH-001	10	2021-06-01 16:17:35	2021-06-01 16:17:35

Figure 25: Sample query from received details

Go back

Next

Refresh

SELECT * FROM receivedetails rd:

Book

IDNo	RRNo	BookID	BookDesc	Qty	DateCreated	DateUpdate
4	RR-00002	978-1111127831	Mathematics for Information Technology / 1st edition / Alfred Basta, Stepha...	5	2021-06-02 18:33:57	2021-06-02 18:33:57
5	RR-00002	978-1250014450	Steve Jobs: The Man Who Thought Different: A Biography	5	2021-06-02 18:33:57	2021-06-02 18:33:57
1	RR-00001	978-1842657881	Fundamentals of Information Technology / Second Edition / S. Kavitha Mur...	10	2021-06-01 18:34:08	2021-06-01 18:34:08
6	RR-00003	978-8470756610	Genealogies of Art, or the History of Art as Visual Art (FONDACION JUAN) / ...	3	2021-06-02 18:46:39	2021-06-02 18:46:39
7	RR-00003	978-9718711064	History of the Filipino People / Online version / Teodoro Agoncillo / Malaya ...	3	2021-06-02 18:46:39	2021-06-02 18:46:39

Figure 26: Sample query from receive header

Go back

Next

Refresh

SELECT * FROM receiveheader rh;

Book

	IDNo	RRNo	VendorID	VendorName	VendorAddress	DTReceive	DateCreated	DateUpdate
▶	1	RR-00001	V-00001	National BookStore	Linketkai Mall, Lapasan, Cagayan de Oro City	2021-06-01	2021-06-01 18:30:24	2021-06-01 18:30:24
	2	RR-00002	V-00002	Rex Book Store Inc.	DDD Holdings Inc. Building, J. Serifa St, Cagayan de Oro	2021-06-02	2021-06-02 18:33:57	2021-06-02 18:33:57
	3	RR-00003	V-00003	United Bookstore	#88 RN Abejuela St, Cagayan de Oro	2021-06-02	2021-06-02 18:46:39	2021-06-02 18:46:39

Figure 27: Sample query from user registration



Go back



Next



Refresh

SELECT * FROM userreg ul;



Book

	IDNo	 UserID	UserName	Password	UserType
▶	1	1	2021-0-00001	Password1234	Borrower
	3	2	2021-0-00002	Pass123	Librarian
	4	3	2021-0-00003	secret@123	Admin

Figure 28: Sample query from vendor list

  		SELECT * FROM vendor vl;				
 Book						
IDNo	VendorID	VendorName	Address	ContactNo	DateCreated	DateUpdate
1	V-00001	National BookStore	Limketkai Mall, Lapasan, Cagayan de Oro City	088-881-7837	2021-06-01 16:27:14	2021-06-01 16:27:14
2	V-00002	Rex Book Store Inc.	DDD Holdings Inc. Building, J. Seriña St, Cagayan de Oro	(088) 858 6775	2021-06-01 16:30:07	2021-06-01 16:30:07
3	V-00003	United Bookstore	#88 RN Abejuela St, Cagayan de Oro	088) 851 5283	2021-06-01 16:31:16	2021-06-01 16:31:16
4	V-00004	C&E Bookshop	Velez St, cor J.R. Borja St, Cagayan de Oro	(08822) 722094	2021-06-01 16:32:22	2021-06-01 16:32:22
5	V-00005	KSD Christian Bookstore	220 Vamenta Blvd, Carmen, Cagayan de Oro	(088) 957 4016	2021-06-01 16:33:07	2021-06-01 16:33:07

APPENDIX B

GANNT CHART FOR LIBRARY MANAGEMENT SYSTEM

Table 19: Schedule of activities of the Dream Team Group

DREAM TEAM SCHEDULE OF ACTIVITIES											
MONTH	February		March		April		May		June		
DAY	10-18	19-28	5-15	16-31	15-18	19-28	12-15	16-28	1-10	17	18
ACTIVITY											
Small group discussion for the proposal of the project											
Waiting for the approval											
Small group discussion for the final proposal											
Consolidation of gathered information											
Planning for the Proccess											
Planning for UI											
Making Flowchart											
Making Diagrams											
Start doing Research											
Meeting/ Group Study of our project											
Development of our program											
Sample Query											
Final Defense											

